Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that the browser sent a query to the DNS server to retrieve the IP address for the website www.yummyrecipesforme.com. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "udp port 53 unreachable". The port noted in the error message is used for the DNS service. The most likely issue is that the DNS server is down, no service is listening on the receiving DNS port, or a firewall is blocking the traffic.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
The incident occurred at 1:24 p.m.. The IT team became aware of the incident when several customers of clients reported that they were unable to access the website www.yummyrecipesforme.com and received a "destination port unreachable" error. The IT department investigated the incident by using a network analyzer tool, tcpdump, to capture and inspect the data packets. The key findings revealed that when UDP packets were sent to the DNS server to resolve the domain name, ICMP packets were returned with the error message "udp port 53 unreachable". The likely cause of the incident is that the DNS server is offline or a firewall configuration is blocking incoming traffic on port 53, preventing users from translating the domain name into an IP address.